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Path planning and tracking for autonomous vehicle based on model predictive control with variable horizon parameters

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Keywords: path planning, tracking, lane-changing, autonomous vehicle, variable horizon parameters, model predictive control

Abstract

Lane-changing (LC) path planning and tracking are the main technologies used to avoid obstacles in autonomous vehicles. To increase safety in path planning and improve tracking performance, an path planning and tracking scheme based on the safe LC area algorithm and the optimized variable horizon parameter model predictive control (MPC) is proposed. First, a judgment method for the safe LC area is presented based on the comprehensive synthesis variable time-headway model, and a quintic polynomial LC path is planned. Second, under different driving speeds and curvature of the path, the weight-adaptive particle swarm optimization algorithm is used to optimize the horizon combination (HC), and the fitting relationship between driving speed, curvature and HC is obtained to calculate the optimized HC in real time, which improves the accuracy of path tracking and reduces the calculation time. Finally, the results show that compared to the other MPC algorithms, the average sum of lateral errors of the proposed adaptive horizon parameter-MPC algorithm is reduced by 21.69%, and the economy and comfort are improved by 13.27% and 13.39%, respectively.

1. Introduction

With the rapid development of technologies such as vehicle control, communication, and artificial intelligence, autonomous driving research has received increasing attention in both academia and industry, and has the advantages of improving driving safety, easing traffic congestion, and improving traffic efficiency [1, 2]. However, the collision problem between vehicles still challenges further development of autonomous driving technology. According to the statistics of the NHTSA, collision accidents caused by driver errors account for more than 92% of all accidents [3, 4]. Therefore, how to safely avoid collisions is the research focus of autonomous driving.

The development of communication technology between vehicles makes information sharing technology a reality [6], which can plan a safe lane-changing (LC) path considering more surrounding vehicles (SVs). Communication between vehicles in complex driving environments has been widely studied [7–9], which has improved the safety and efficiency of LC to some extent. The characteristics of the researches are summarized in table 1. The LC decision method considering SV information shows great research potential and application prospects. Not only does it overcome the problems of low reliability and low decision accuracy, but it also improves the global security of LC path planning of the ego vehicle (EV).

Currently, parameter-based methods are the most widely used path-planning methods. The parametric method generates multiple smooth curves in the driving direction of the vehicle, and obtains the optimal planned path between the curves. In early planning, lines and circular arc were used to plan the shortest path between global waypoints [10], but the discontinuity of the path curvature would lead to

Table 1. Comparison of vehicle communication technologies.

	Method	Scenario	Limitation
[7]	Nonlinear integrated parameter model and delayed feedback control	connected automated vehicles	Without considering echo characteristics of communication
[8]	Improved distributed MPC controller	Two lanes and three vehicles	The heterogeneity of vehicles affects cooperative communication
[9]	Scenario and weight adaptive MPC	Two lanes and five vehicles	Without considering driving style and communication delay

the decrease in tracking accuracy, reducing the vehicle's driving comfort. In contrast, the Sigmoid function is widely used for path planning, owing to its continuous smoothness and variable parameters. By changing the slope of the Sigmoid function, trajectories with different curvature parameters can be obtained to satisfy the driving requirements of LC and overtaking [11, 12]. Simultaneously, the Sigmoid function can also be used as a boundary condition to plan a safe LC area and serve as the security constraint condition of the decision level [13]. Because of the advantages of simple control and easy generation of complex smooth curves, the Bezier curve is often used to fit the segmental path on the basis of segmental path planning, to form a continuous smooth LC path [14, 15]. Polynomial path is widely used in LC path planning because of its simple parameters and order-determining performance. For example, a cubic polynomial path guarantees smooth lateral velocity, a quintic polynomial path guarantees smooth curvature, and a seventh-degree polynomial path guarantees smooth lateral acceleration. Moreover, this method exhibits high operational efficiency [16]. Li *et al* [17] proposed an LC algorithm that could generate a quintic polynomial function path to optimize the overall benefits of autonomous vehicles and SVs. Lim *et al* [18] combined sampling and a numerical optimization approach and generated multiple lateral movements based on a quintic polynomial, enhancing the ability to adapt to various driving situations. Although there are many types of paths used in existing path planning research, no study jointly optimizes LC path generation and model predictive control (MPC) horizon parameters based on both curvature and speed.

Several controllers have been studied to accurately track the path, including robust control [19, 20], fuzzy control [21, 22], PID control [23], sliding mode control [24], and MPC [25]. Among these methods, due to its good performance in dealing with non-linear systems, MPC has been widely used to solve nonlinear vehicle dynamics tracking control [26]. MPC not only provides an easy way to incorporate environmental constraints and dynamic vehicle constraints into optimization problems, it also allows for easy assigning weights to targets to achieve the desired performance [27–29]. However, there are also some drawbacks in MPC. First, the computational complexity of MPC increases as the prediction and control horizon expand, making it difficult to satisfy real-time computing requirements. Second, fixed horizon MPC is prone to cause calculation delays, which poses a threat to driving safety. Therefore, to achieve a dynamic balance between 'real-time computing' and 'optimal control', adaptive prediction and control horizon have become key technologies.

The most important parameters in the MPC algorithm are the prediction horizon and the control horizon, which form a horizon combination (HC). The constant horizon combined MPC can only accurately track the path under certain path curvature and vehicle speed. Therefore, as the curvature and vehicle speed change, the HC needs to be dynamically adjusted to achieve the best control effect. Some research results have been obtained for the dynamic horizon adjustment. Liu *et al* [30] utilized longitudinal velocity and front wheel steering angle as fuzzy variables, and proposed a method for obtaining the prediction and control horizon through fuzzy reasoning. However, the selection of fuzzy reasoning rules is prone to be influenced by the driver's style, and the curvature changes of the road are not taken into account. Li *et al* [31] experimentally obtained the optimal prediction and control horizon parameters at different vehicle speeds, and an adaptive HC parameter path tracking controller was designed. However, only limited optimal results can be obtained using experimental methods, and the influence of changes in road curvature on HC has not been considered. Therefore, it is necessary to study the combined effect of road curvature and vehicle speed on HC, and to obtain the optimal HC by combining with optimization algorithms. Therefore, it is necessary to introduce an optimization method into the solution process to obtain the optimal HC when the vehicle speed and curvature change.

To enhance the adaptability of MPC controllers to complex environments, many scholars have conducted relevant research on the dynamic parameters of MPC controllers. The particle swarm optimization (PSO) method is used for the offline optimization of the horizon parameters of the MPC controller to increase the adaptability of vehicles to the driving environment [32, 33]. The PSO is a heuristic

algorithm for solving non-linear and discontinuous optimization problems and finds an optimal solution by iterative improvement of a candidate solution based on a cost function. The main advantages of PSO are its simplicity, computational efficiency, and optimization properties [34]. Given the strict latency requirements of onboard automotive controllers (<10 ms), PSO's low complexity is advantageous. Tao and Kim [35] proposed a dual-population PSO algorithm using a random disturbance strategy, which is superior to the existing PSO algorithm in terms of the path tracking accuracy and running time. Li *et al* [36] used the linear attenuation weight method to update the inertia weight of PSO, and the PSO-optimized MPC controller maintained a path tracking error within 0.28 m. Wang *et al* [37] adopted the variable inertia weight PSO algorithm to optimize the MPC prediction horizon and improve the accuracy of path tracking in different curvature environments. However, little attention has been paid to the optimization of HC. At the same time, there are few reports on HC optimization that consider the influence of vehicle speed and road curvature. Therefore, it is necessary to study a real-time adjustment method for HC.

By summarizing the above issues, a path planning and tracking control scheme is proposed considering the driving state of SVs, offline optimization, and an online real-time path tracking control method. The innovations of this study are summarized below.

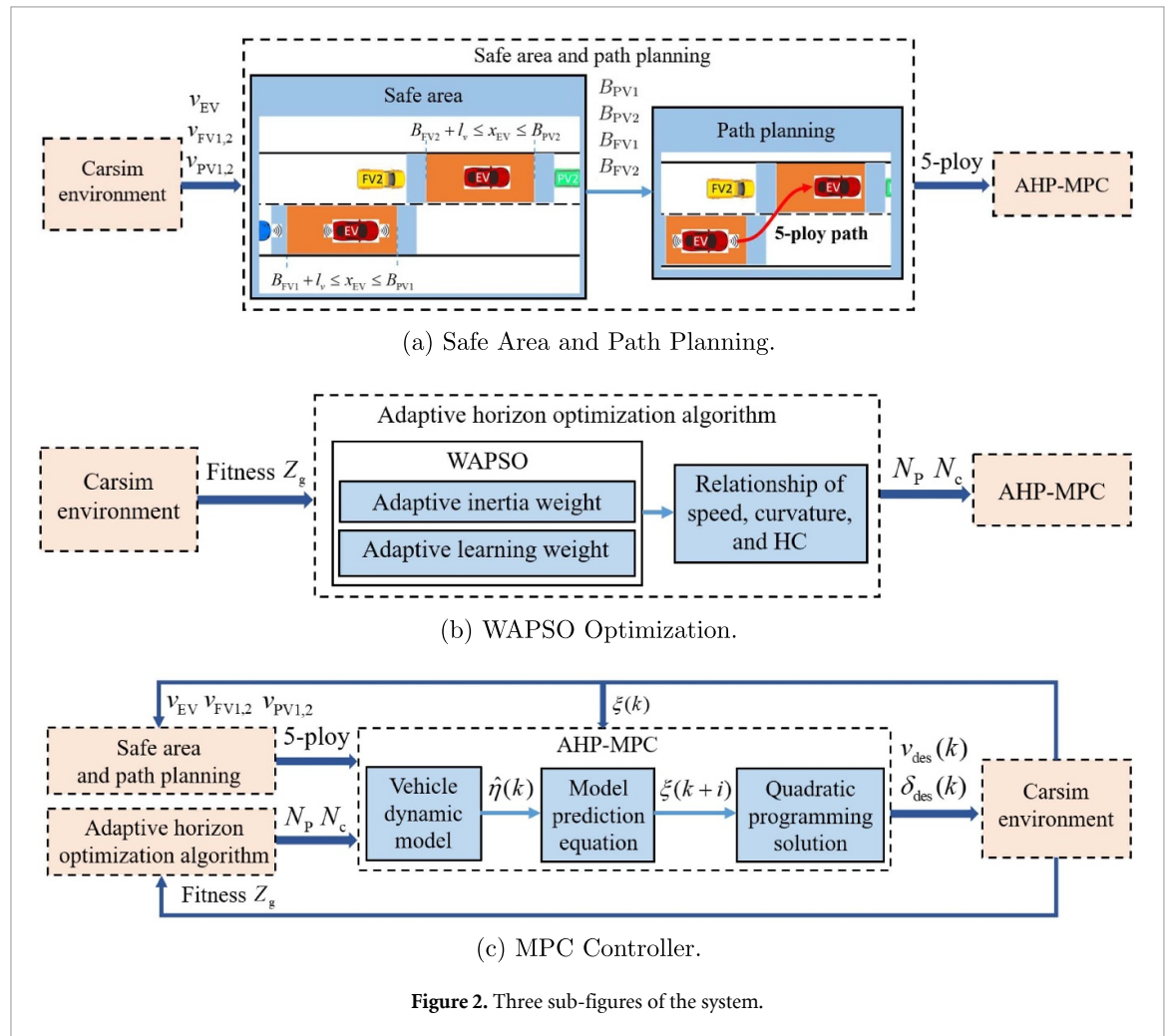
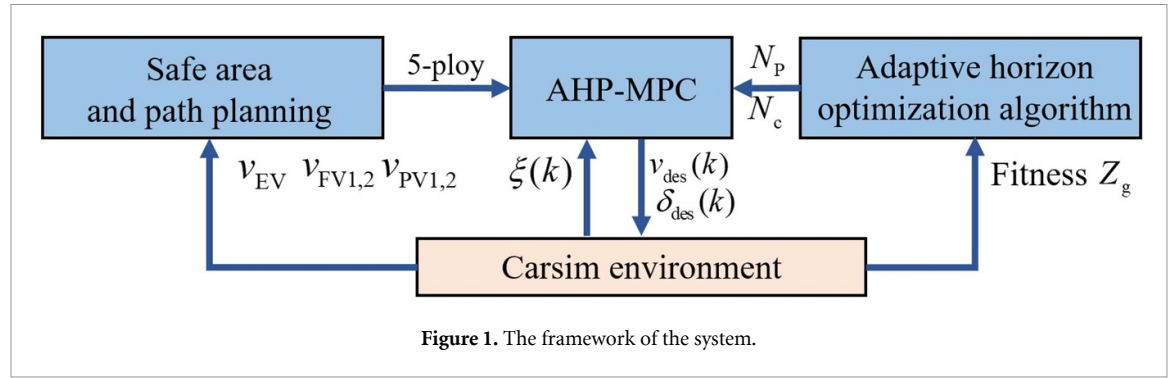
- (1) Based on the continuous synthetic variable time headway model (CSVTH), a method is presented to determine the safe LC area considering the speed and acceleration of the SVs. Unlike Daoud, Jiang, Ammour, *et al* [11–13], a path selection method based on the minimum sum of curvature is proposed, which effectively reduces the lateral acceleration of the vehicle during LC and improves comfort.
- (2) Considering the horizon parameters of the MPC controller as particles, the weight-adaptive PSO (WAPSO) is proposed to adaptively update the weight of the particle velocity. Unlike Tao, Li, Wang, *et al*. [35–37], it can update both the inertia and learning weight based on historical fitness, effectively improving the local and global optimization ability of the particles. Based on the WAPSO, the optimal HC can be obtained for different speeds and road curvatures.
- (3) Unlike Feng and Wang [7], we calculate the optimal HC in real-time by fitting the relationship between vehicle speed, curvature, and HC, thus continuously computing the HC and improving precision. Additionally, an adaptive horizon parameter MPC (AHP-MPC) is designed to boost the path tracking accuracy and decrease the computation time.

The remainder of this paper is organized as follows. Section 2 briefly introduces the overall structure of the study. In section 3, combined with the CSVTH model, the safety area required by the EV to change lanes is calculated, and the corresponding judgment conditions for safe LC are proposed. In section 4, the vehicle dynamics model is established, and the MPC controller is designed. The WAPSO method, which can adaptively update the weight of the particle velocity, is detailed in section 5. The simulations and results are discussed in section 6. The final section concludes this paper.

2. Overall structure of the system

This study adopts a hierarchical simulation structure, which is mainly divided into four parts: path planning layer, MPC controller layer, PSO parameter optimization layer, and Carsim environment layer, as shown in figure 1.

- (1) The block of safe area and path planning is presented in figure 2(a). The output of SV and ego vehicle from the Carsim are $v_{PV1,2}$, $v_{FV1,2}$ and v_{EV} , respectively. Based on the $v_{PV1,2}$, $v_{FV1,2}$, v_{EV} and CSVTH, the safe areas are obtained. With the help of safe areas, the 5-ploy path is planned.
- (2) The block of adaptive horizon optimization algorithm acquires the fitness value from Carsim and combines the WAPSO algorithm to optimize N_p and N_c , obtaining formulas (20) and (21) in the paper under different speeds and curvatures.
- (3) The block of AHP-MPC obtains the key parameters of the 5-ploy path, N_p , N_c and $\xi(k)$ performs a calculation of the control quantity. Finally, it outputs the angle of the front wheel and the vehicle speed to the Carsim, thereby achieving vehicle control.
- (4) The block of Carsim environment exhibits the vehicle driving and path following.



3. Safe lane-changing area

3.1. Safe lane-changing area judgment

To realize collision-free LC of the EV, it is very important to determine the starting point and end point of the LC. The width of the LC path is determined by the width of the road, whereas the length of the path must be determined by considering the position and speed of the SVs. The common LC scenario on a highway is shown in figure 2. We only consider the four vehicles closest to the EV; the vehicles in front (PV1) and behind the EV (FV1) in the current lane, and the vehicles in front (PV2) and behind the EV (FV2) in the target lane. The connected vehicles can accurately and quantitatively obtain the position and speed information of the SVs, calculate the positions of the vehicles at the future moment, and then obtain the safe LC area of the EV. The LC process is briefly described as follows.

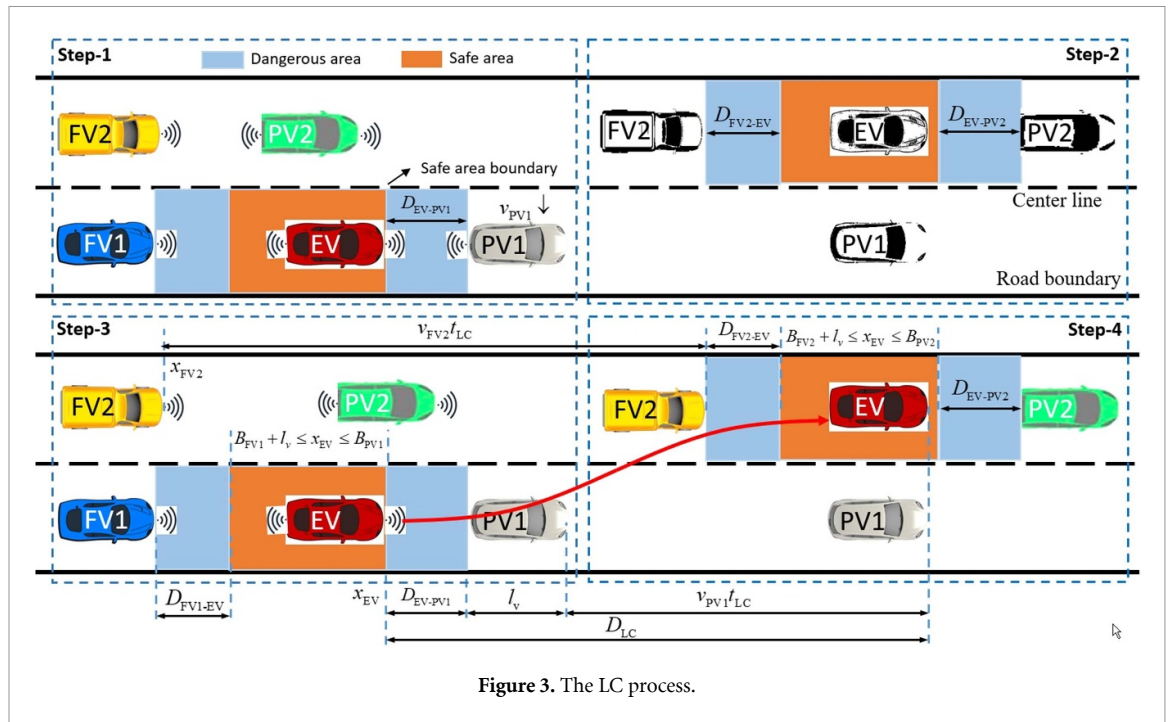


Table 2. Explanation of the abbreviations in figure 3.

Name	Method	Scenario
EV	ego vehicle	The controlled vehicle in this work
SV	surrounding vehicle	The vehicles surrounding the EV
PV1	preceding vehicle 1	The vehicle in front of the EV in the current lane
PV2	preceding vehicle 2	The vehicle in front of the EV in the target lane
FV1	following vehicle 1	The vehicle following the EV in the current lane
FV2	following vehicle 2	The vehicle following the EV in the target lane

In the dotted box step-1 in figure 3, PV1 in the current lane slows down and the EV reaches the dangerous blue area of PV1; therefore, the LC function of the EV is activated to avoid collision with PV1.

In the dotted box step-2, the EV calculates the orange safety area of the adjacent lane based on the speeds and positions of the SVs obtained from the Internet of vehicles. The virtual vehicles in black and white indicate that all the vehicles are likely to be after the LC.

In the dotted box step-3, the EV calculates the length required for LC according to the safety area and plans the LC path.

In the dotted box step-4, the EV completes the path tracking and enters the safe LC area.

The explanations of the variables in figure 3 are listed in table 2.

In a safe area study, although the Gipps algorithm of the existing study considers the braking capacity of different vehicles [38], the braking capability is susceptible to external interference, resulting in the failure of the safe distance calculation. The core problem in safe area calculation is the calculation of the time headway between vehicles (THBV). In this study, CSVTH, the previous research achievement is used to calculate the safe area [39], which reduces the interference of uncertainty of the parameters in the safe distance calculation and improves the reliability of the safe distance calculation. In the first step, the blue dangerous areas D_{FV1-EV} and D_{EV-PV1} of both FV1 and PV1 trigger the calculation of the safe area LC of the EV. Taking D_{EV-PV1} in Step-3 as an example, the calculation formula of CSVTH is (1) considering both velocity and acceleration differences between EV and PV,

$$T_{\text{CSVTH}} = t_0 + c_v v_{\text{EV}} - sg_1(D_r) c_v v_{\text{PV1}} - sg_2(D_r) c_a a_{\text{PV1}} \quad (1)$$

where t_0 is the safe time constant; c_v is the weight of the difference between v_{EV} and v_{PV} ; $sg_1(D_r)$ and $sg_2(D_r)$ are the transition functions based on the sigmoid, all of which are explained in our previous research [45]; v_{PV} is the PV speed; a_{PV} and c_a are the acceleration and its weight, respectively.

Based on (1), equation (2) calculates the THBV corresponding to the blue danger area in figure 3,

$$\begin{bmatrix} T_{\text{CSVTH}}^{\text{EV-PV1}} \\ T_{\text{CSVTH}}^{\text{EV-PV2}} \\ T_{\text{CSVTH}}^{\text{FV1-EV}} \\ T_{\text{CSVTH}}^{\text{FV2-EV}} \end{bmatrix} = \begin{bmatrix} t_0 \\ c_v \\ -sg_1(D_r) \\ -sg_2(D_r) \end{bmatrix}^\top \begin{bmatrix} 1 & 1 & 1 & 1 \\ v_{\text{EV}} & v_{\text{EV}} & v_{\text{FV1}} & v_{\text{FV2}} \\ v_{\text{PV1}} & v_{\text{PV2}} & v_{\text{EV}} & v_{\text{EV}} \\ a_{\text{PV1}} & a_{\text{PV2}} & a_{\text{EV}} & a_{\text{EV}} \end{bmatrix} \quad (2)$$

where $T_{\text{CSVTH}}^{\text{EV-PV1}}$, $T_{\text{CSVTH}}^{\text{EV-PV2}}$, $T_{\text{CSVTH}}^{\text{FV1-EV}}$, and $T_{\text{CSVTH}}^{\text{FV2-EV}}$ represents the THBV between EV and PV1, EV and PV2, FV1 and EV, and FV2 and EV, respectively.

The corresponding distance between vehicles, that is, the blue danger area, is expressed as (3).

$$\begin{cases} D_{\text{EV-PV1}} = v_{\text{EV}} T_{\text{CSVTH}}^{\text{EV-PV1}} \\ D_{\text{EV-PV2}} = v_{\text{EV}} T_{\text{CSVTH}}^{\text{EV-PV2}} \\ D_{\text{FV1-EV}} = v_{\text{FV1}} T_{\text{CSVTH}}^{\text{FV1-EV}} \\ D_{\text{FV2-EV}} = v_{\text{FV2}} T_{\text{CSVTH}}^{\text{FV2-EV}} \end{cases} \quad (3)$$

To accurately calculate the longitudinal LC distance based on the CSVTH, the starting point of the longitudinal LC distance is the first point of the path, and the end point is the point with the same coordinates as the front of the PV1 after the EV completes the LC as shown in Step-4. Equation (4) calculates the longitudinal LC distance.

$$\begin{cases} D_{\text{EV-PV1}} + l_v + v_{\text{PV1}} t_{\text{LC}} = v_{\text{EV}} t_{\text{LC}} \\ t_{\text{LC}} = \frac{D_{\text{EV-PV1}} + 2l_v}{v_{\text{EV}} - v_{\text{PV1}}} \\ D_{\text{LC}} = v_{\text{EV}} t_{\text{LC}} \end{cases} \quad (4)$$

where, l_v is the EV length, t_{LC} is the time required for the LC, D_{LC} is the LC distance.

According to figure 3, equation (5) calculates the safety area of the EV before and after the LC.

$$\begin{bmatrix} B_{\text{PV1}} \\ B_{\text{PV2}} \\ B_{\text{FV1}} \\ B_{\text{FV2}} \end{bmatrix} = \begin{bmatrix} 1 & t_{\text{LC}} & -1 & -v_{\text{EV}} \\ 1 & t_{\text{LC}} & -1 & -v_{\text{EV}} \\ 1 & t_{\text{LC}} & 0 & v_{\text{FV1}} \\ 1 & t_{\text{LC}} & 0 & v_{\text{FV2}} \end{bmatrix} \cdot \begin{bmatrix} x_{\text{PV1}} & x_{\text{PV2}} & x_{\text{FV1}} & x_{\text{FV2}} \\ v_{\text{PV1}} & v_{\text{PV2}} & v_{\text{FV1}} & v_{\text{FV2}} \\ l_v & l_v & l_v & l_v \\ T_{\text{CSVTH}}^{\text{EV-PV1}} & T_{\text{CSVTH}}^{\text{EV-PV2}} & T_{\text{CSVTH}}^{\text{FV1-EV}} & T_{\text{CSVTH}}^{\text{FV2-EV}} \end{bmatrix} \quad (5)$$

Considering the length of the EV, the judgment condition of the safe area before and after LC can be represented by (6).

$$\begin{cases} B_{\text{FV2}} + l_v \leq x_{\text{EV}} \leq B_{\text{PV2}} \\ B_{\text{FV1}} + l_v \leq x_{\text{EV}} \leq B_{\text{PV1}} \end{cases} \quad (6)$$

When the front bumper coordinate of the EV (x_{EV}) satisfies this judgment condition, as shown in Step-3 and Step-4 in figure 3, the EV will not collide with the SVs before and after the LC, thereby achieving a safe LC.

3.2. path planning

A polynomial curve of appropriate order can ensure that motion parameters such as velocity, acceleration, and jerk are continuous and smooth. The order of the polynomial curve is determined by the boundary conditions, and the trajectories are usually polynomials of degrees three, five, or seven. Because the vehicle boundary conditions generally contain the initial and terminal states of the vehicle, the number of states of the vehicle is usually even, which means that polynomials of odd order are required to solve the problem. Owing to the existence of continuous high-order derivatives, the quintic polynomial (5-ploy) satisfies the vehicle state constraints; therefore, it is widely used in path planning [16–18]. Taking the EV (60 km h^{-1}) overtakes the PV (30 km h^{-1}) as an example, the four LC paths are plotted in figure 4. The sum of curvatures of the four paths are summarized in table 3. Compared with the Sigmoid curve, the tanh curve, and the Bezier curve used in other studies [12, 15], the curvature of

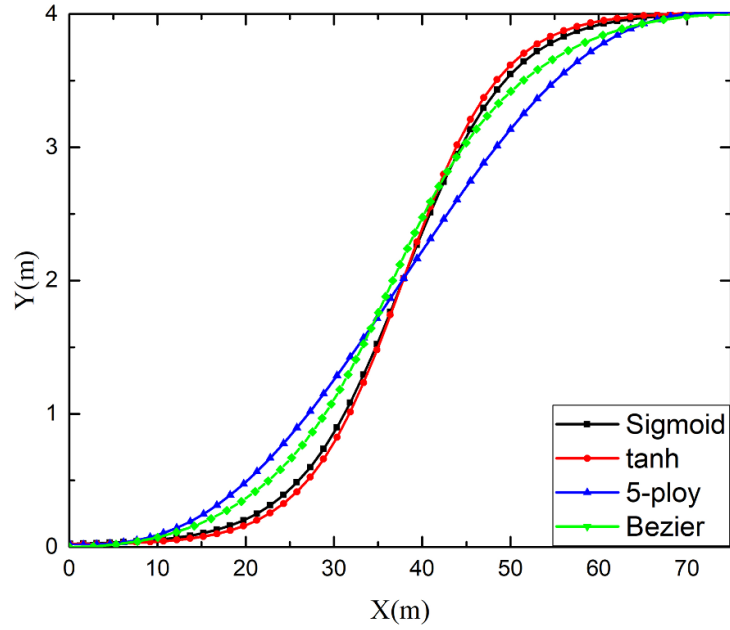


Figure 4. Comparison diagram of the four paths.

Table 3. Sum of the curvatures.

	Sigmoid	tanh	5-ploy	Bezier
Sum of curvatures	0.2134	0.2322	0.1303	0.2263

the 5-ploy is the smallest. Therefore, the EV had the best lateral comfort and economy when tracking the 5-ploy path.

The expression for the 5-ploy is written as (7):

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 \quad (7)$$

where a_i is the coefficient of the 5-ploy, which is determined by the starting and terminal states of the vehicle.

At the beginning and end of the LC, the vehicle state satisfies (8) :

$$\begin{cases} y(0) = 0, y'(0) = 0, y''(0) = 0 \\ y(D_{LC}) = W_{road}, y'(x_{end}) = 0, y''(x_{end}) = 0 \end{cases} \quad (8)$$

where W_{road} is the road width and x_{end} is the end point of the 5-ploy path.

Substituting (8) into (7), the path formula can be obtained as (9):

$$y(x) = 10W_{road} \left(\frac{x}{D_{LC}} \right)^3 - 15W_{road} \left(\frac{x}{D_{LC}} \right)^4 + 6W_{road} \left(\frac{x}{D_{LC}} \right)^5 \quad (9)$$

Because the width of the LC path is equal to the width of the road, the LC interval is determined by D_{LC} . Therefore, a unique LC path is obtained.

4. path tracking control

4.1. Dynamic model

The vehicle dynamics model is used as a prediction model for the MPC. To accurately describe the vehicle dynamics process and reduce the computational load of the control algorithm, the following assumptions are made.

- (1) To reduce the influence of the road simulation environment on the controller, this paper adopts a planar road to construct the road, without considering the banked road scenario.

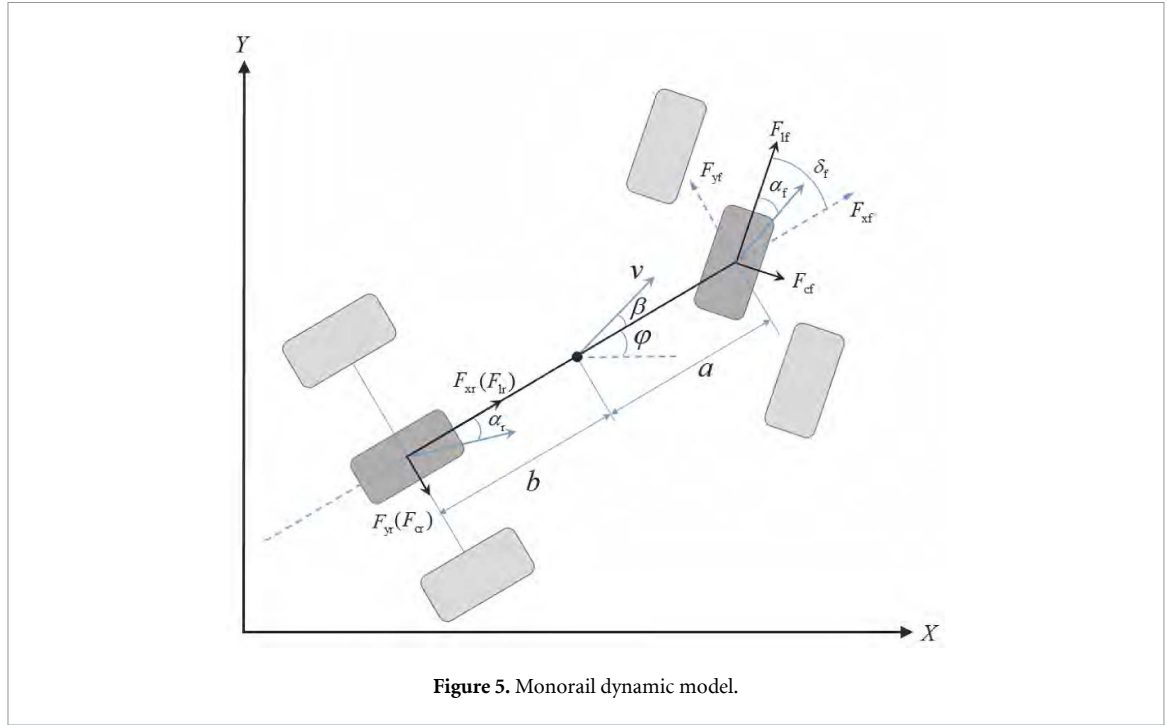


Figure 5. Monorail dynamic model.

- (2) The characteristics of the pure sideward tires is only considered, ignoring the coupling relationship between lateral and longitudinal tire forces.
- (3) It is considered that the wheel track is negligible in relation to the turning radius.

The three-DOF vehicle monorail dynamics model is established as shown in figure 5, and the vehicle state equation based on this dynamic model is shown in (10). The establishment of the dynamic model takes into account the small-angle hypothesis of the tires, and in the simulation, the slip angle of the vehicle satisfies the small-angle hypothesis. Therefore, the lateral and longitudinal forces have the linear relationships with the slip angle and slip rate, respectively.

$$\begin{cases} \ddot{y} = -\dot{x}\dot{\varphi} + \frac{2}{m} \left[C_{cf} \left(\delta_f - \frac{\dot{y} + a\dot{\varphi}}{\dot{x}} \right) + C_{cr} \frac{b\dot{\varphi} - \dot{y}}{\dot{x}} \right] \\ \ddot{x} = \dot{y}\dot{\varphi} + \frac{2}{m} \left[C_{lf}\delta_f - C_{cf} \left(\delta_f - \frac{\dot{y} + a\dot{\varphi}}{\dot{x}} \right) \delta_f + C_{lf}\delta_r \right] \\ \dot{\varphi} = \dot{\varphi} \\ I_z \ddot{\varphi} = 2 \left[aC_{cf} \left(\delta_f - \frac{\dot{y} + a\dot{\varphi}}{\dot{x}} \right) - bC_{cr} \frac{b\dot{\varphi} - \dot{y}}{\dot{x}} \right] \\ \dot{Y} = \dot{x} \sin \varphi + \dot{y} \cos \varphi \\ \dot{X} = \dot{x} \cos \varphi - \dot{y} \sin \varphi \end{cases} \quad (10)$$

where $\ddot{y}$ and $\ddot{x}$ are the lateral and longitudinal accelerations of the vehicle in the vehicle coordinate system, $\dot{x}$ and $\dot{y}$ are the longitudinal and lateral velocity of the vehicle in the vehicle coordinate system, $\dot{\varphi}$ is the yaw angle speed, φ is the yaw angle, m is the weight of the autonomous vehicle, C_{cf} and C_{cr} are the lateral stiffness of the tires, δ_f is the front wheel angle, a and b are the distances from the center of mass to the front and rear axes of a vehicle, I_z is the inertia moment of the body around the Z axis. $\dot{X}$ and $\dot{Y}$ are the longitudinal and lateral velocity of the vehicle in the geodetic coordinate system.

The (10) is represented as a state space $\dot{\xi} = f(\xi, u)$, and u denotes δ_f . The state quantity is selected as $\xi = [\dot{y}, \dot{x}, \varphi, \dot{\varphi}, Y, X]^T$, and the yaw angle and lateral displacement are selected as the output of the system $\eta = [\varphi \quad Y]^T$. The state space equation can be written as (11):

$$\begin{cases} \dot{\xi} = f(\xi, u) \\ \eta = C \cdot \xi \end{cases} \quad (11)$$

$$\text{where } C = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

In the $k + 1$ control cycle, based on the already derived vehicle dynamics model, the corresponding discrete form of (11) is written as (12):

$$\begin{cases} \xi(k+1) = \xi(k) + Tf(\xi(k), u(k)) = F(\xi(k), u(k)) \\ \eta(k) = C \cdot \xi(k) \end{cases} \quad (12)$$

where T denotes the sampling period.

To generate the increment of the controller, a new state variable $\hat{\xi}(k) = \begin{bmatrix} \xi(k) \\ u(k-1) \end{bmatrix}$ is defined. The $\hat{\xi}(k+1)$ can be written as:

$$\begin{cases} \hat{\xi}(k+1) = \hat{A}_k \hat{\xi}(k) + \hat{B}_k u(k) + \hat{d}(k) \\ \hat{\eta}(k) = \hat{C} \hat{\xi}(k) \end{cases} \quad (13)$$

where $\hat{A}_k$ and $\hat{B}_k$ are the coefficients of the state equation; $\hat{d}(k)$ is the error of the state equation; $\hat{C} = [C \quad 0]$.

4.2. Objective function

In the process of tracking path control of the AV, the accuracy and economic performance of vehicles must be taken as optimization goals. The smaller the lateral error and yaw angle are, the higher the tracking accuracy of the path will be. The economic performance refers to the control increment of the front-wheel steering angle. The smaller the change in the front-wheel steering angle, the lower the energy consumption. Based on the above analysis, the objective function is written as (14).

$$J(\hat{\eta}(k), \Delta u(k)) = \sum_{i=1}^{N_p} \|\hat{\eta}(k+i) - \hat{\eta}_{\text{ref}}(k+i)\|_Q^2 + \sum_{i=0}^{N_c-1} \|\Delta u(k+i)\|_R^2 + \rho \varepsilon^2 \quad (14)$$

where N_p is the prediction horizon, N_c is the control horizon, Q and R are the output weighting matrix and control quantity increment weighting matrix, $\rho \varepsilon^2$ is the relaxation term.

In the simulation, the R is set as 10^5 based on the existing research results [31, 41, 43], and the Q is determined by the experiments. When the value of Q is 10^3 , our AHP-MPC controller can satisfy the experimental requirements.

The output of the vehicle controller must consider the physical constraints of the actual actuators. These are summarized as the control quantity and control increment constraints. The path tracking controller should solve the optimization problem (15) in each control cycle.

$$\begin{aligned} & \min(J(\hat{\eta}(k), \Delta u(k))) \\ & \text{s.t.} \begin{cases} U_{\min} \leq U_{\text{ctrl}} \leq U_{\max} \\ \Delta U_{\min} \leq \Delta U_{\text{ctrl}} \leq \Delta U_{\max} \\ -\mu_{\min} g \leq a_y \leq \mu_{\max} g \\ \varphi_{\min} \leq \varphi \leq \varphi_{\max} \\ Y_{\min} \leq Y \leq Y_{\max} \\ \varepsilon > 0 \end{cases} \end{aligned} \quad (15)$$

where U_{ctrl} denotes the control variable, and $(U_{\min}, U_{\max})$ restricts the range of the U_{ctrl} ; ΔU_{ctrl} denotes the increment of the U_{ctrl} , and $(\Delta U_{\min}, \Delta U_{\max})$ restricts the range of the ΔU_{ctrl} ; $(\mu_{\min} g, \mu_{\max} g)$ restricts the range of the a_y ; $(\varphi_{\min}, \varphi_{\max})$ denotes the range of the yaw angle φ ; $(Y_{\min}, Y_{\max})$ denotes the range of the lateral position of the vehicle; ε is the relax factor.

After repeated iterations in the above processes, the tracking control is realized.

5. Adaptive horizon optimization algorithm

In a highway environment, the prediction horizon N_p and the control horizon N_c of the MPC algorithm have significant impacts on the real-time performance and accuracy of the tracking. A set of optimal HC values of the N_p and N_c can better balance the accuracy of tracking and the computational efficiency on the highway. Therefore, it is necessary to determine the optimal horizon parameters and construct an MPC control algorithm with adaptive horizon parameters.

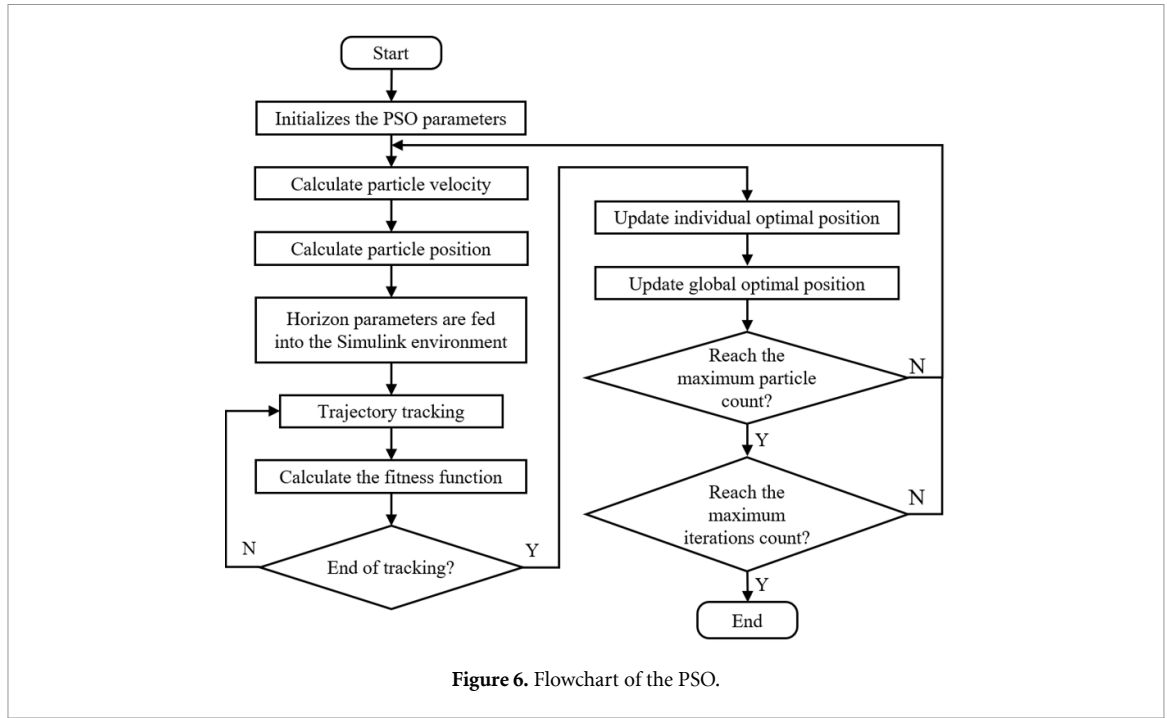


Figure 6. Flowchart of the PSO.

5.1. Particle swarm optimization

The PSO is an iterative algorithm commonly used in optimization calculations, and its working process is described below.

First, the initialized particles are sent to the optimization object to obtain the corresponding output result, which is the fitness value. Subsequently, the velocity and position of the particle swarm are calculated according to the velocity calculation formula and the position calculation formula, and the global optimal position and individual optimal position of the particle are updated. Finally, the updated particle value is sent to the optimization object to obtain a new fitness value. If the termination condition is satisfied, the optimal global solution is the output. Otherwise, the loop is continued. The velocity and position update formulas are shown in (16), and the flowchart is presented in figure 6.

$$\begin{cases} v_{ij}(t+1) = \omega v_{ij}(t) + c_1 r_1(t) (p_{ij}(t) - x_{ij}(t)) + c_2 r_2(t) (p_{gj}(t) - x_{ij}(t)) \\ x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t+1) \end{cases} \quad (16)$$

where ω is the inertia weight, $v_{ij}(t)$ is the particle speed, $x_{ij}(t)$ is the position of the particle; c_1 and c_2 are the positive learning weights, $r_1(t)$ and $r_2(t)$ are two independent random functions in the range $[0, 1]$.

5.2. Weight adaptive PSO

5.2.1. Adaptive inertia weight

The traditional PSO algorithm has the disadvantages of a low accuracy and easy divergence. Existing research uses the linear variable inertia weight method, whose inertia weight is updated linearly with iteration episodes [40, 41]. We name this approach hard-update PSO (HUPSO). Because the change in inertia weight does not consider the relationship between current fitness and global fitness, the fitness function and inertia weight do not match. This will mislead the particles to the wrong position, resulting in a local optimum, or optimization failure which seriously affects the computational efficiency of the optimization algorithm.

From (16), it becomes clear that the inertia weight ω and the learning weights c_1 and c_2 directly influence the velocity of the particles throughout the iterations, subsequently affecting their searching abilities and convergence speed. During the initial stages of the iteration, a larger ω , combined with a larger c_1 and a smaller c_2 , promotes global exploration of the solution space, effectively preventing premature convergence. As the iterations progress, to improve the local convergence abilities of the algorithm, the inertia weight should gradually decrease, whereas a smaller c_1 and a larger c_2 help the convergence of the particles towards the optimal solution.

Based on the analysis presented above, this paper proposes a method for updating the inertia weight ω by considering the decrease in the fitness function. When the fitness function becomes larger, the

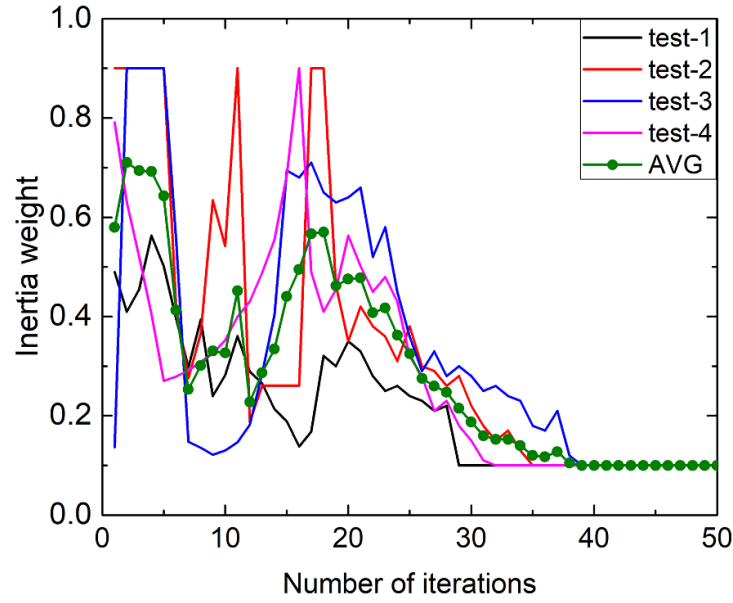


Figure 7. The changing process of inertia weight ω .

performance of the particle worsens, and the original particle velocity should be reduced to find a better solution in the global space. When the fitness function decreases, the performance of the particles improves, and the weight should be retained to accelerate the particle optimization speed. The improved formula for calculating ω is as (17):

$$\begin{cases} \omega_{\min} + (\omega_{\max} - \omega_{\min}) \left(\frac{z - z_{\min}}{z_{\text{avg}} - z_{\min}} \right), & z \leq z_{\text{avg}} \\ \omega_{\max}, & z > z_{\text{avg}} \end{cases} \quad (17)$$

where $\omega_{\min}$ and $\omega_{\max}$ are the minimum and maximum values of the inertia weight, z is the real-time fitness, z_{avg} and $z_{\min}$ are the average and minimum values of the fitness of all particles in the previous round.

Because the PSO algorithm has the property of a Markov chain, this study only uses the mean and the max–min values of the fitness of the last optimization episode to update the inertia weight, and has nothing to do with the fitness values of other episodes. Therefore, the speed of the location update is improved.

The average fitness of the last episode of the particles is used as the reference point of the current episode, and the inertia weight is adjusted by the ratio of the current particle fitness to the average fitness of the last episode. If the current fitness is greater than the average of the previous episode, the current particle moves away from the optimal point, and the inertia weight needs to be increased to maintain the original velocity direction. If the current fitness is less than the average value of the last episode, it means that the current particle approaches the optimal point, and the weight can be reduced according to (17) to encourage the particle to update speed.

Because the velocity update of each particle episode of particles is related to the minimum and average fitness values of the previous episode, the inertia weight update of each time can match the current fitness function, avoiding the blindness of the inertia weight in HUPSO. We call the improved weight update method soft update, and the corresponding PSO algorithm soft update PSO (SUPSO).

The plotted ω curve is shown in figure 7. The green dot line represents the result obtained by averaging the inertia weight data. At the initial of optimization, the inertia weight randomly varies between the minimum and maximum values, reflecting the process by which particles explore the optimal position. In the later stage of optimization, the inertia weight gradually converges towards the minimum value, indicating the process by which particles explore near the optimal position to avoid local optima.

5.2.2. Adaptive learning weights

The learning weight determines the ability of the current particle to learn the individual and global extremum. The c_1 denotes the ability of the current particle to learn the individual optimal position, and the c_2 denotes the ability of the current particle to learn the global optimal position. They reflect the

communication between the particles. Without these two factors, the particles will only search according to the current speeds and will not have the ability to adjust the speeds adaptively.

With fixed learning weights, it is very likely that optimal local solutions will be found when dealing with complex problems. Therefore, this study changes the learning weight by increasing the number of iterations, and the updated formula is as (18):

$$\begin{cases} c_1 = c_{\max} - (c_{\max} - c_{\min}) \sin(\pi k / (2k_{\max})) \\ c_2 = c_{\min} + (c_{\max} - c_{\min}) \sin(\pi k / (2k_{\max})) \end{cases} \quad (18)$$

where $c_{\max}$ and $c_{\min}$ are the maximum and minimum values of the learning weight, k and $k_{\max}$ are the current and maximum iteration episodes, respectively.

The learning weights change the influence of the individual and global extrema on the position of the particles according to the number of iterations. At the early stage of optimization, the particle is significantly affected by the individual extreme value; therefore, the particle can quickly find a relatively good position. In the middle stage of optimization, some particles are close to the global optimal value, and some continue to search for better position points. At the end of the optimization, when the number of iterations reaches the maximum number of iterations, all particles are close to the global optimal value, so that the particles can achieve better convergence. The PSO algorithm adopted in this study includes an adaptive inertia weight and an adaptive learning weight, which is called WAPSO.

5.3. Comparison results

To verify the effectiveness of the proposed WAPSO algorithm, a tracking experiment is carried out on the same circular path under different vehicle speeds, and the HC optimization process is recorded in detail.

The initial number of particles in the experimental simulation is 20, and the simulation rounds are set to 50. The locations of 20 groups of the HCs are randomly initialized and the constraint relation: $N_p > N_c$ is satisfied. The velocity constraint range is $[-2, 2]$, inertia weight $\omega_{\max}$ and $\omega_{\min}$ are 0.9 and 0.1, and the learned weights $c_{\max}$ and $c_{\min}$ are 0.9 and 0.1. The duration of the path tracking is 4 seconds. In this period, the output of the lateral displacement error is summed at each control time, and the result is returned to the main PSO program as the fitness value Z_g . At the same time, the historical optimal HC is updated with the current HC. Otherwise, it is abandoned. At this point, the optimization process of one particle is completed.

This paper studies LC path tracking in a highway environment; therefore, the speed of the autonomous vehicle is selected as 100 km h^{-1} . The maximum radius of curvature corresponding to the LC path is 900 m. The base PSO (BPSO), HUPSO, SUPSO and WAPSO are selected for the optimization comparison experiments, and each algorithm is repeated 20 times. The experimental results of N_p , N_c , Z_g , and the computation time T_{cons} are shown in figure 8.

In figure 8(a), the N_p based on the BPSO algorithm converges to 12 after 24 iterations, and the corresponding lateral error sum is 3.71 m in figure 8(c). The N_p values based on values HUPSO, SUPSO, and WAPSO algorithms converge to 11 after 21, 17, and 12 iterations respectively, and the corresponding lateral error sums are 3.65 m, 3.55 m, and 3.51 m respectively. In figure 8(b), N_c based on the BPSO and HUPSO algorithms converges to 5 and 4 after 30 iterations, respectively, and the N_c based on the SUPSO and the WAPSO algorithms both converge to 3 after 18 and 13 iterations, respectively. From the above results, we can see that the N_p and the N_c based on the BPSO are larger than those of the three modified PSO algorithms. This indicates that the HC is trapped at the local optimum point during the optimization process based on the BPSO algorithm. The three improved PSO algorithms can not only jump out of the local optimum, but also search the global optimum faster.

In figure 8(c), the optimization time of each episode based on the WAPSO algorithm is reduced by 66.58 s, 30.51 s and 13.84 s compared with the BPSO, HUPSO and SUPSO algorithms, respectively. The WAPSO algorithm can effectively reduce the invalid optimization time caused by particle random search and improve the efficiency of particle optimization. Meanwhile, the standard deviation (S.D.) of the optimization time used by the WAPSO algorithm has also decreased by 9.66%, 7.12% and 3.04% in table 4, respectively, indicating that the improved PSO algorithm is more stable in terms of optimization time. Because the WAPSO algorithm can obtain the optimal HC with the fastest speed, in figure 8(d), the global optimal fitness values of each episode based on the WAPSO algorithm are 5.39%, 3.81% and 1.68% lower than those of the BPSO, HUPSO and SUPSO, respectively, which effectively improves the accuracy of trajectory tracking. Meanwhile, the fitness value of the WAPSO algorithm has the lowest S.D. in table 5, indicating that the stability of the particle search process has been further enhanced.

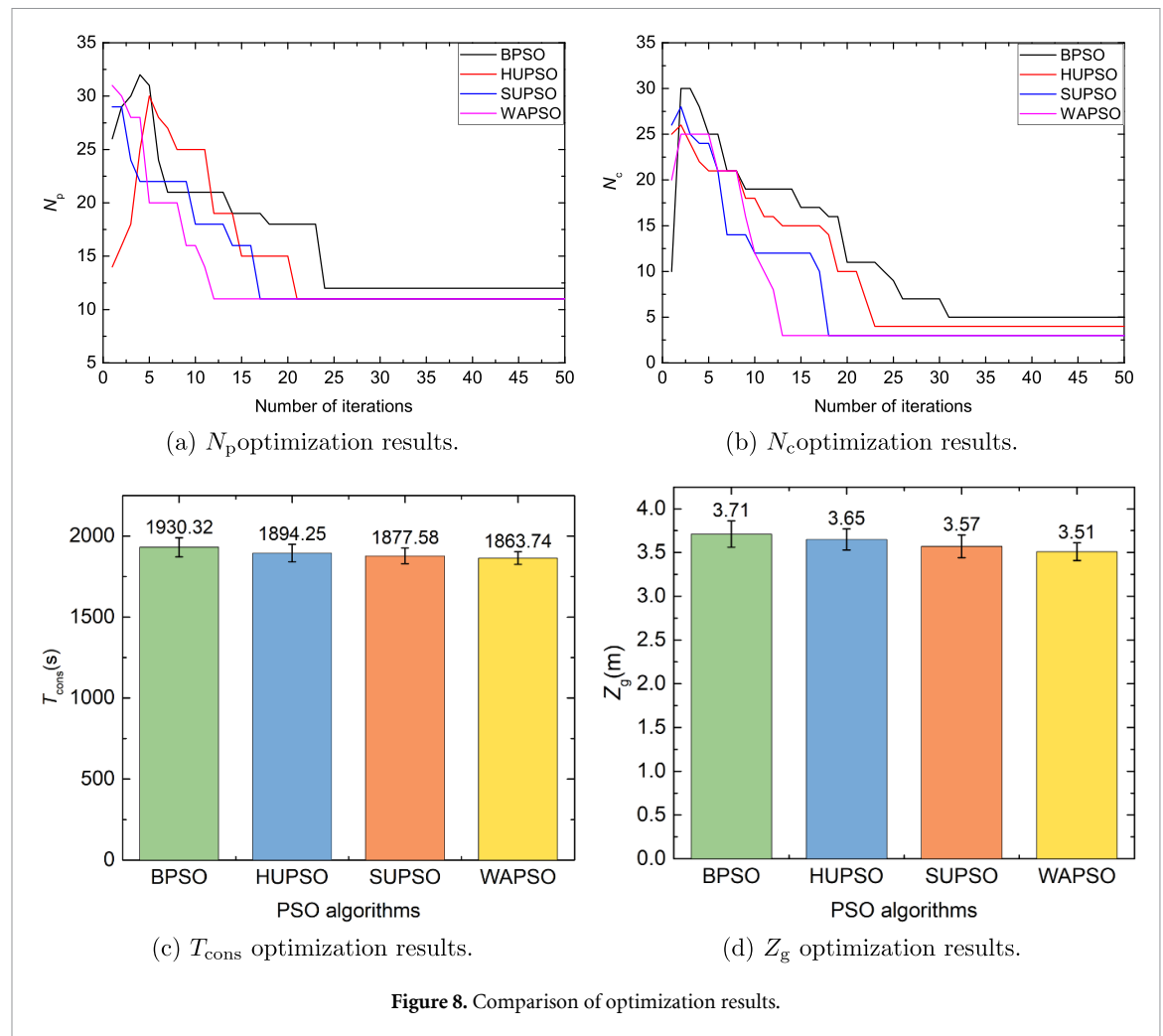


Figure 8. Comparison of optimization results.

Table 4. T_{cons} results.

	Average	S.D.
BPSO	1930.32	58.15
HUPSO	1894.25	54.25
SUPSO	1877.58	48.62
WAPSO	1863.74	39.14

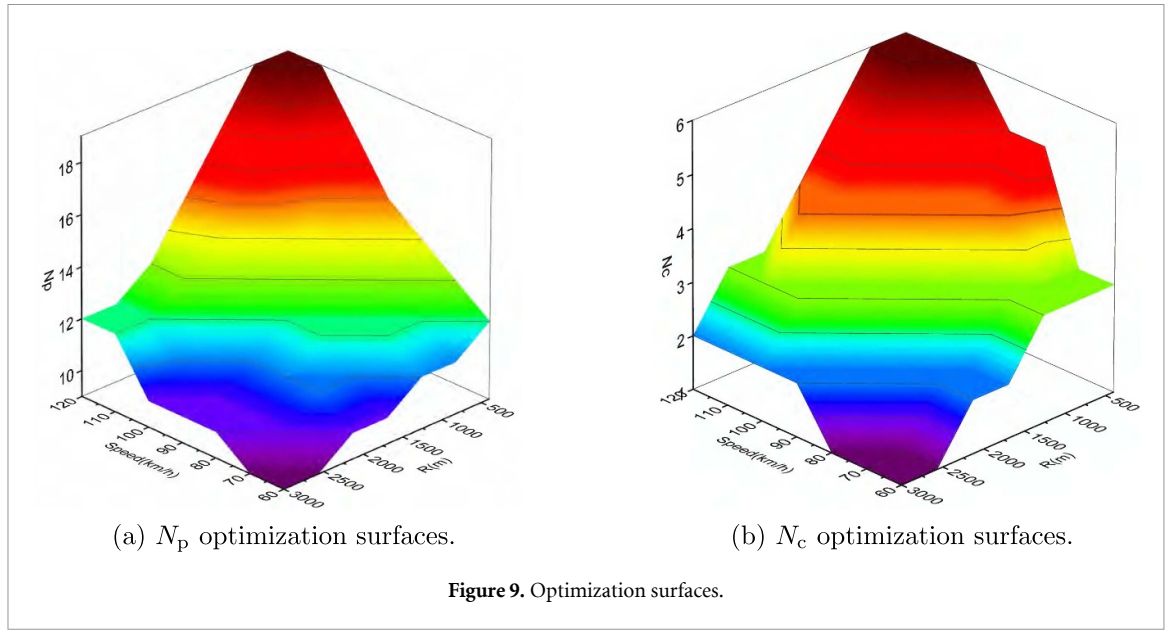
Table 5. Z_g results.

	Average	S.D.
BPSO	3.71	0.15
HUPSO	3.66	0.12
SUPSO	3.57	0.13
WAPSO	3.51	0.10

5.4. Relationship between vehicle speed, curvature, and HC

In the path-tracking process, the controller selects the optimal HC according to the speed and curvature. Therefore, the fit relationship between the vehicle speed, curvature, and the HC is established. When the vehicle tracks the path, the controller can calculate the optimal HC in real-time according to its own speed and the curvature of the path to improve the tracking accuracy and calculation efficiency.

However, in the actual path tracking process, the optimization process requires considerable computing time; therefore, it is not easy to realize a real-time optimization process. Based on the above reasons, this study proposes a the technical scheme for offline optimization and an online solution.



First, at different vehicle speeds, the possible path curvatures are calculated offline using (19), and the corresponding optimal HC is solved using the proposed WAPSO for each curvature.

$$R = \frac{v_{\max}^2}{K^2 A_f \mu g} \quad (19)$$

where R is the radius of the path, K is the safety factor, 0.74 through the experiment; A_f is the safety factor (0.8), limiting tire friction saturation, μ is the coefficient of friction 0.8, g is the acceleration due to gravity.

Second, the radius is gradually increased and, at each vehicle speed, the HC with the minimum tracking error is found by the WAPSO algorithm.

Third, the relationship between the vehicle velocity, curvature, and optimal HC is fitted, and the optimal HC is a function of the vehicle velocity and curvature. The optimization surfaces are shown in figure 9.

By adopting polynomial-2D fitting, the HC fitting results in figure 9 are as follows: (20) and (21):

$$\begin{aligned} Z_{N_p} = & 643.09 - 39.12v_{EV} + 0.95v_{EV}^2 - 0.01v_{EV}^3 \\ & + 6.72 \times 10^{-5}v_{EV}^4 - 1.56 \times 10^{-7}v_{EV}^5 \\ & + 0.002R - 6.15 \times 10^{-6}R^2 \end{aligned} \quad (20)$$

$$\begin{aligned} Z_{N_c} = & 373.35 - 21.39v_{EV} + 0.49v_{EV}^2 - 0.01v_{EV}^3 \\ & + 3.02 \times 10^{-5}v_{EV}^4 - 6.58 \times 10^{-8}v_{EV}^5 \\ & - 0.001R - 1.39 \times 10^{-6}R^2 \end{aligned} \quad (21)$$

where, Z_{N_p} and Z_{N_c} are the values of N_p and N_c .

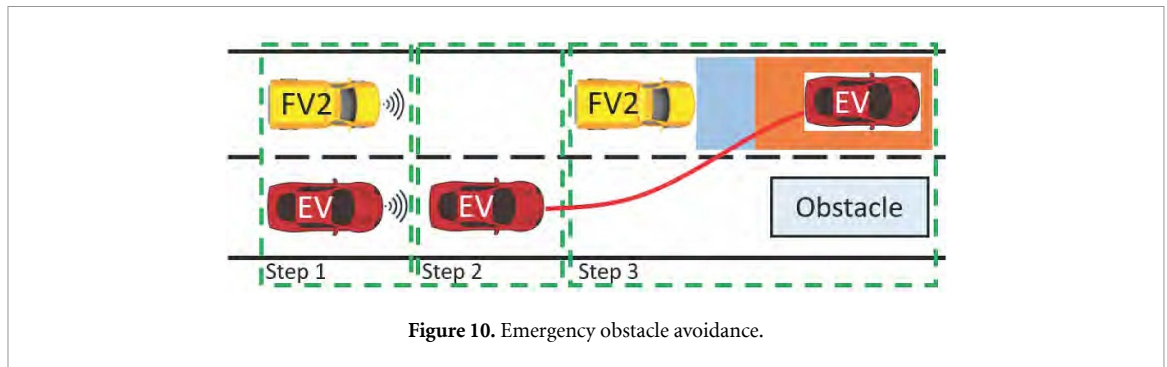
Finally, based on (20) and (21), the optimal HC combination can be calculated according to the real-time planned path curvature and vehicle speed to achieve accurate path tracking. The MPC path tracking method that uses (20) and (21) to calculate the optimal HC in real-time is called AHP-MPC.

6. Results and discussion

6.1. Emergency obstacle avoidance

In this section, we study the difference between the AHP-MPC and several fixed-parameter MPCs in the emergency obstacle avoidance at different speeds. When stationary obstacles are present in the lane, the vehicle must track the planned safe path in time and accurately, and ensure the lateral stability.

This part of the scenario is shown in figure 10 can be divided into three stages: In the step 1, a stationary obstacle appears in front of the EV when traveling at a speed of 25 m s^{-1} in the lane. FV2 in the adjacent lane travels normally at a slower speed than the EV. In the step 2, after calculating the safe LC



area, the EV determines that a safe LC area appears in front of the FV2, and plans the LC path according to the safe LC area. In the step 3, the EV follows the path to avoid obstacles.

The three most commonly used constant HC (CHC) are compared with those of AHP-MPC. They are $(N_p = 20, N_c = 2)$, $(N_p = 13, N_c = 6)$, and $(N_p = 22, N_c = 10)$, and they are named CHC-1 [13], CHC-2 [44], and CHC-3 [45], respectively. The experiments are compared in terms of tracking error, yaw angle error, lateral acceleration, and calculation time. The experimental results are shown in figure 11.

In figure 11(a), the EV enters the start state of the LC after the simulation starts, and the curvature of the path changes most significantly. As shown in figure 11(h), the prediction horizon increases from 11 to 16, and the control horizon increases from 1 to 4. This is because when the curvature increases, the vehicle must adjust the angle of the front wheel in time to quickly reduce the tracking error. At this time, using a small prediction horizon cannot obtain a sufficiently large accumulation of prediction errors, which will lead to the front wheel angle solved by MPC not satisfying the tracking requirements. Therefore, the prediction horizon must be increased to ensure that the MPC solver has sufficient error input.

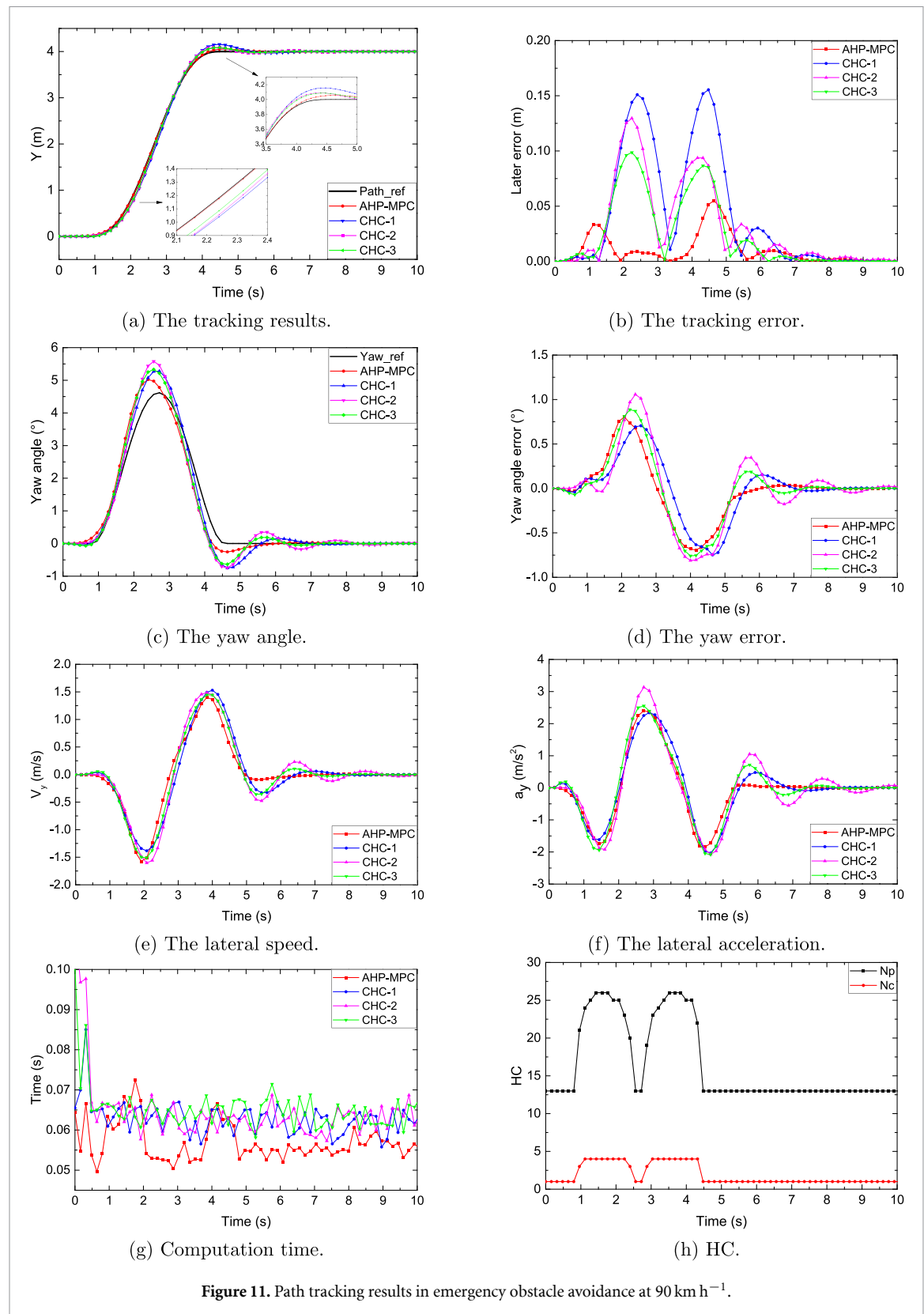
The control horizon will also increase with curvature. In the MPC, an appropriate increase in the control horizon can improve the solving accuracy of the solver to some extent. Therefore, the HC should increase synchronically with the curvature. Owing to the increase in the HC, the AHP-MPC changes the angle of the front wheel earlier than the other three algorithms; therefore, the error (red curve) appears earlier in figure 11(b). At the same time, the speeds and accelerations in figures 11(e) and (f) also show an early change. Although there is a lateral tracking error, it can increase the steering time of the vehicle and reduce the speed of change in the front wheel angle, thus reducing the lateral impact and improving the comfort level.

When simulated to approximately 2–3.5 s, the EV enters a stable LC state. In this interval, the 5-poly is characterized by a small curvature and approximates a straight line. This is because when tracking a straight line, there is essentially no difference between large HC and small HC, and both can satisfy the tracking error requirements. Therefore, in this interval, the MPC only needs a small HC to accurately track the path. This can be observed from figure 11(g) that the calculation time of AHP-MPC in this period is the least.

After 3.5 s, the LC entered the attitude adjustment stage, and the curvature of the path increased again, similar to the starting state of the LC; therefore, the HC increased again. The local magnification diagram shown in figure 11(a), the LC path based on the AHP-MPC is closest to the reference path, with an overshoot of 5.46%, whereas the overshoot of the other three methods is 15.41%, 8.86% and 8.57%, respectively. At the same time, this can be observed in figures 11(c) and (d) that the yaw angle based on the AHP-MPC changes more smoothly. This is because, in the attitude adjustment stage, the HC is constantly adjusted with the change in curvature, ensuring that the AHP-MPC-based front wheel accurately tracks the path in a timely manner.

To clearly compare the differences between the four methods, we summarize the lateral error, yaw angle error, acceleration, and the simulation time in table 6.

It can be clearly seen from table 6 that the sum of lateral tracking errors of the AHP-MPC method is the least, which is only 25.15%, 33.59% and 41.97% of the other three methods, respectively. The sum of lateral errors based on the AHP-MPC method has the smallest S.D. (± 0.10), indicating that the sum of lateral errors is sensitive to the changes in HC parameters. In addition, compared with the other three methods, the yaw angle based on the AHP-MPC is reduced by 5.6%, 26.21% and 13.71%, respectively. Due to the continuous change of the HC, the lateral acceleration based on the AHP-MPC method is significantly reduced, which means that the comfort has been significantly improved. Due to the automatic



reduction of the HC in the flat track stage, the calculation time based on the AHP-MPC method is the least in the whole LC process, which effectively reduces the calculation load of the on-board controller. All the S.D. results indicate that the simulation results have good consistency.

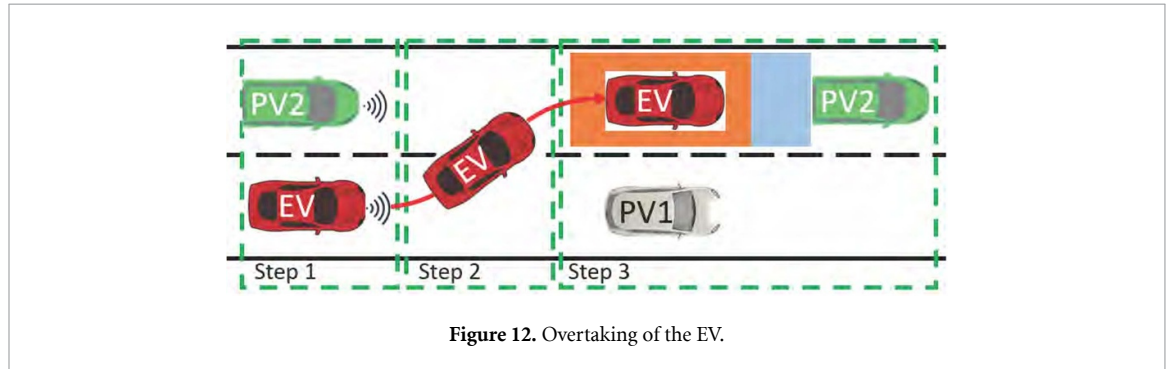
6.2. Overtaking

6.2.1. Scenario 1

Overtaking is the most common scenario for EV driving on highways. Slow vehicles in front of the EV will seriously affect the traffic efficiency of the road; therefore, an accurate and fast overtaking maneuver

Table 6. Indicators of the obstacle avoidance.

Indicators	AHP-MPC	CHC-1	CHC-2	CHC-3
$\sum \Delta Y_{err} (m)$	1.28 ± 0.10	5.09 ± 0.12	3.81 ± 0.12	3.05 ± 0.11
$\sum \Delta \varphi_{err} (^{\circ})$	24.86 ± 0.55	26.33 ± 0.60	33.69 ± 0.62	28.81 ± 0.59
$\sum a_y (ms^{-2})$	75.18 ± 0.35	84.18 ± 0.39	100.69 ± 0.38	88.88 ± 0.41
$\sum t_e (s)$	7.81 ± 0.13	8.42 ± 0.14	8.54 ± 0.17	8.66 ± 0.17

**Figure 12.** Overtaking of the EV.

is one of the essential functions of autonomous vehicles, and it is necessary to study them carefully and deeply. In this section, we perform a simulation analysis using the example of an EV overtaking PV1 (16.67 m s^{-1}) at 25 m s^{-1} .

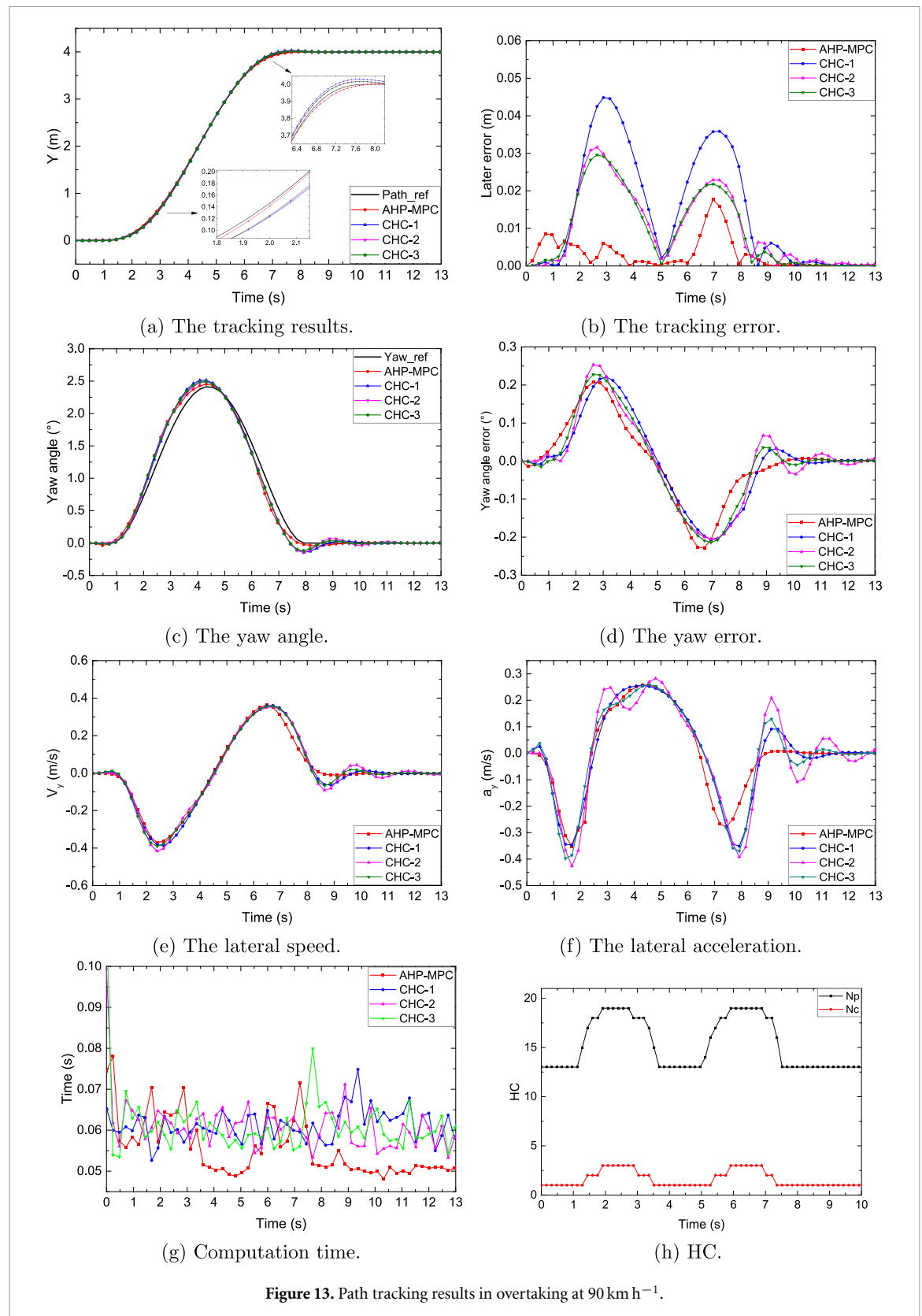
This scenario is illustrated in figure 12, which can be divided into three stages. In the step 1, when the EV is driving at 25 m s^{-1} in the lane, there is a slow-moving vehicle PV1 in front of it, and the PV2 in the adjacent lane is driving faster. In the step 2, after calculating the safe LC area, the EV is judged to have a safe LC area behind the PV2, and the LC path is planned. In the step 3, the EV starts tracking and completes the overtaking.

It can be clearly noticed from the two local magnification diagrams in figure 13(a) that the tracking path of the AHP-MPC method is closest to the reference path, no matter at the beginning or the end of the LC; therefore, the tracking error is also the smallest among the four methods. At the end of the LC, the vehicle must adjust its attitude in time to accurately control the vehicle running in the middle of the lane. The EV with optimized HC results adjusted the angle of the front wheel in time according to the change in curvature; therefore, the overshoot of the vehicle's yaw angle is only 4.03%. The overshoots of the other three methods are 11.73%, 14.3% and 14.38%, respectively. For the same reason, in figures 13(e) and (f), the transverse velocity and acceleration of the EV also obtain the smoothest transition curve, effectively reducing the risk of rolling. The speed of the other three methods overshoot twice, five, and three times, respectively, before reaching a stable state, and the adjustment time lags behind that of the AHP-MPC method by 4.83 s, 2.41 s, and 1.36 s, respectively.

Compared to the emergency obstacle avoidance scenario, the relative speed between the two vehicles in the dynamic overtaking scenario is much lower because the vehicle in front also has a certain speed. Consequently, the required LC distance also increases significantly. As shown in figure 13(g), the corresponding simulation time is increased by approximately 3 s compared to the obstacle avoidance scenario.

The sums of the lateral errors, yaw angle errors, acceleration, and simulation time are summarized in table 7.

It can be clearly seen from table 7 that the sum of the lateral tracking errors of the AHP-MPC method is the lowest. In addition, the sum of the yaw angle error based on the AHP-MPC method is reduced by 10.77%, 15.75% and 13.35%, respectively, compared to the other three methods. Because of the continuous change in the HC, the lateral acceleration based on the AHP-MPC method is significantly reduced. Owing to the automatic reduction of HC in the flat track stage, the calculation time based on the AHP-MPC method also shows advantages in the entire LC process, which is reduced by 9.47%, 9.13% and 9.05% compared with other methods, thereby improving the controller calculation efficiency. It can be clearly seen from table 7 that the AHP-MPC algorithm also maintained the minimum S.D. Among the four indicators, the S.D. of the AHP-MPC algorithm is respectively 10.74%, 14.83%, 14.86% and 17.56% lower than the average of the other three CHC methods, proving the consistency of the AHP-MPC.



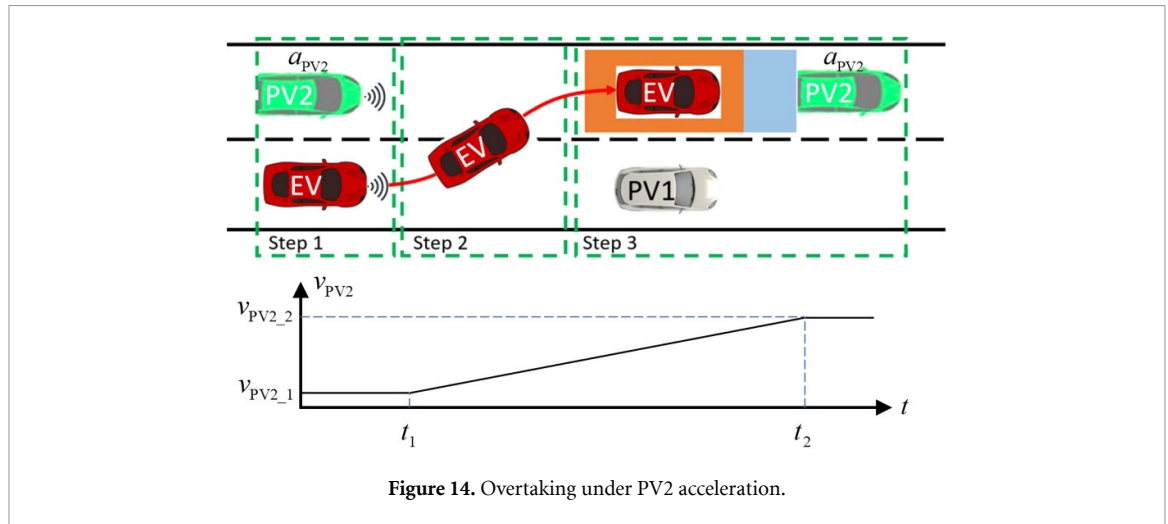
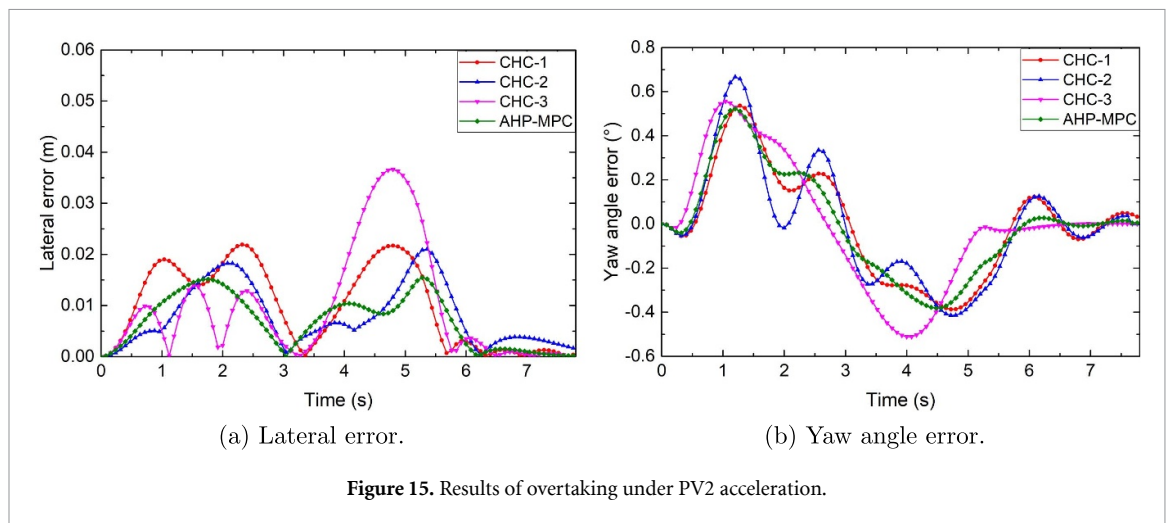
6.2.2. Scenario 2

To simulate a more realistic overtaking scenario, we conducted an accelerating scenario to simulate a more realistic overtaking and test robustness, as shown in figures 14 and 15. The PV2 in the adjacent lane accelerates from t_1 to t_2 at an acceleration of 1.38 m s^{-2} , and the vehicle speed ranges from 16.67 m s^{-1} to 25 m s^{-1} .

As can be seen from table 8, the sum of the absolute values of the lateral errors using the AHP-MPC method is reduced by 14.28%, 5.01% and 13.4%, respectively, compared to the other three methods. The

Table 7. Indicators of the overtaking.

Indicators	AHP-MPC	CHC-1	CHC-2	CHC-3
$\sum \Delta Y_{err} (m)$	1.21 ± 0.08	2.36 ± 0.09	1.59 ± 0.09	1.53 ± 0.10
$\sum \Delta \varphi_{err} (^{\circ})$	11.18 ± 0.26	12.53 ± 0.31	13.27 ± 0.33	12.64 ± 0.28
$\sum a_y (ms^{-2})$	18.56 ± 0.34	20.97 ± 0.38	23.85 ± 0.42	21.42 ± 0.40
$\sum t_e (s)$	9.75 ± 0.12	10.77 ± 0.13	10.73 ± 0.15	10.72 ± 0.16

**Figure 14.** Overtaking under PV2 acceleration.**Figure 15.** Results of overtaking under PV2 acceleration.**Table 8.** Indicators of the accelerating scenario.

Indicators	AHP-MPC	CHC-1	CHC-2	CHC-3
$\sum \Delta Y_{err} (m)$	1.68 ± 0.07	1.96 ± 0.09	1.77 ± 0.08	1.94 ± 0.08
$\sum \Delta \varphi_{err} (^{\circ})$	17.08 ± 0.51	18.64 ± 0.58	19.11 ± 0.61	18.57 ± 0.57

sum of the absolute values of the yaw angle errors is reduced by 8.37%, 10.62% and 8.02%, respectively. Compared with the existing data in the paper, the AHP-MPC algorithm has reduced the lateral errors and yaw angle errors, demonstrating the stability of the AHP-MPC in various overtaking scenarios.

6.3. Compare with other variable horizon methods

In this section, we mainly compare the AHP-MPC with three recent variable-horizon MPCs. Method 1 optimized the HC based on the vehicle speed and designed HC selection in the form of a piecewise function, denoted VH-1 [31]. In Method 2, a neural network is used to predict the HC, denoted VH-2 [43], based on the vehicle speed and the friction coefficient of the road. In method 3, based on vehicle speed and curvature, the optimal prediction horizon is obtained by looking up a table, denoted VH-3

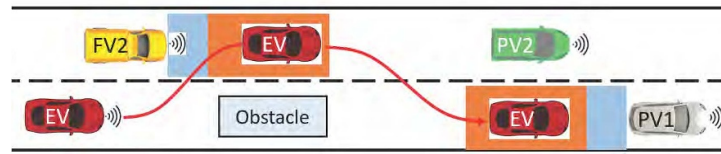


Figure 16. Comprehensive scenario of overtaking.

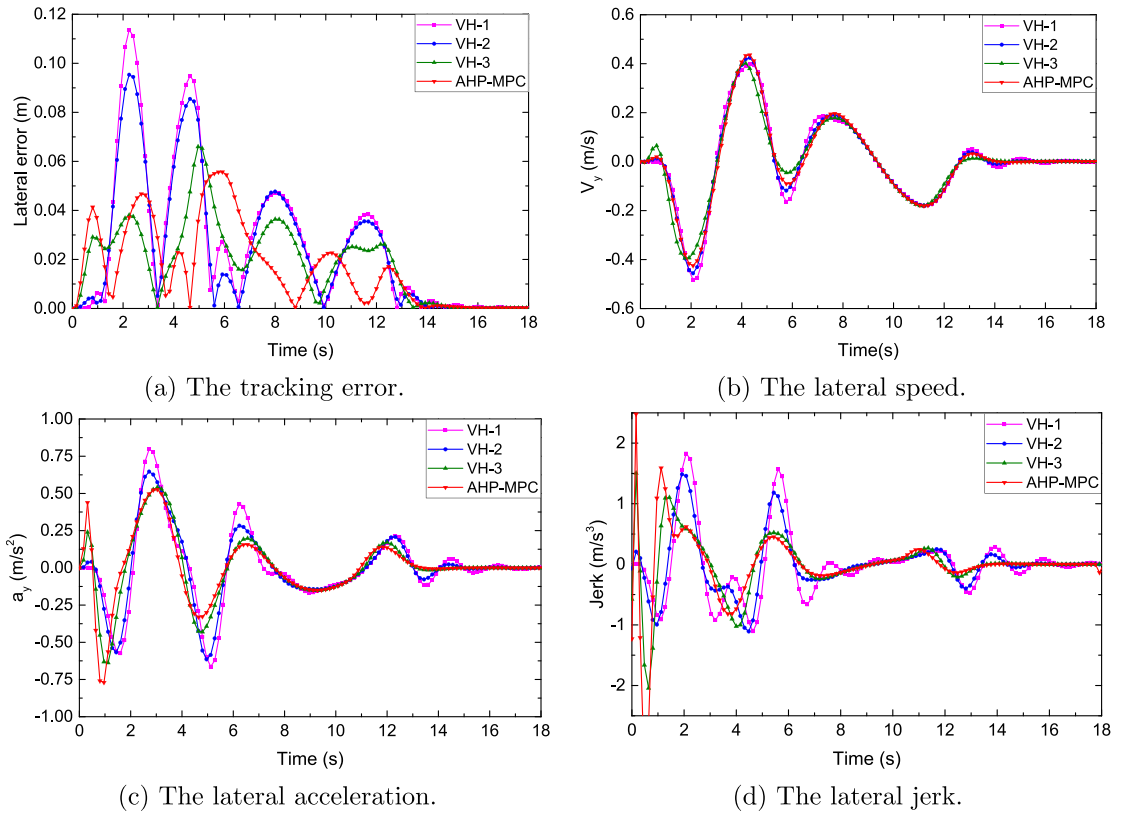


Figure 17. Results of the comprehensive scenario of overtaking.

[37]. The three methods are same with this paper, using discrete MPC based on dynamic model for path tracking, and Simulink/Carsim co-simulation environment for experiments.

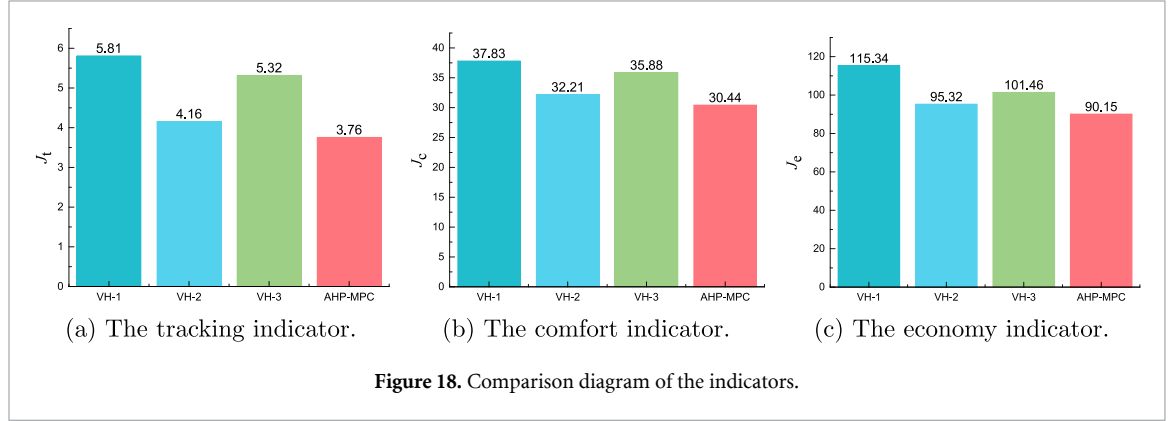
To capture the differences among the algorithm performances, a comprehensive scenario is used to simulate, as shown in figure 16. The EV travels at a speed of 30 m s^{-1} , whereas the SVs also travel at different speeds. First, an EV overtakes a stationary obstacle that appears on the road. Second, after passing the stationary obstacle, there is a slow vehicle PV2 in front of the EV, and the EV plans the LC path by determining the safe area of the LC. Finally, the EV overtakes the slow vehicle to complete the path tracking.

We performed a comparative analysis of three aspects of lateral tracking error, lateral acceleration, and lateral impact, and used tracking performance, comfort and economy to illustrate the advantages of the AHP-MPC algorithm. The experimental results are shown in figure 17.

In figure 17(a), although the VH-1 method can simultaneously change the HC according to the vehicle speed, the optimal solution of the HC cannot be guaranteed using the experimental method combined with the piecewise function. Therefore, the tracking error is maximum. From the comparison of lateral tracking errors, the fluctuation amplitude of the error is significantly lower than that of the other three methods. Because the other three methods cannot integrate the vehicle speed and curvature to obtain the optimized HC and improve the tracking performance, therefore, it can be concluded that the results of path tracking are relatively sensitive to the changing method of HC. Simultaneously, discontinuous HC changes, as shown in figures 17(c) and (d), lead to the maximum lateral acceleration

Table 9. Indicators of the comprehensive scene.

Indicators	AHP-MPC	VH-1	VH-2	VH-3
$\sum \Delta Y_{\text{err}} $ (m)	3.76 ± 0.08	5.81 ± 0.09	4.16 ± 0.10	5.32 ± 0.10
$\sum a_y $ (ms^{-2})	30.44 ± 0.32	37.83 ± 0.35	32.21 ± 0.41	35.88 ± 0.40
$\sum j_y $ (ms^{-3})	59.71 ± 0.89	77.51 ± 0.92	61.31 ± 0.94	65.58 ± 0.96



and jerk which seriously affect the comfort and economy of driving. As shown in figures 17(a) and (d), the VH-2 method does not consider the influence of road curvature on the prediction horizon. When the curvature changes, the optimal prediction horizon cannot be used to control the vehicle. Therefore, the tracking accuracy is lower than that of the AHP-MPC. Although the VH-3 method considers the influence of vehicle speed and curvature on the HC and optimizes the prediction horizon, it does not optimize the control horizon, affecting the improvement of MPC tracking performance.

Based on the existing research results [46–48], the following three formulas (22)–(24) are used to represent the three indexes of tracking performance, comfort and economy.

$$J_t = \sum_{i=1}^n |\Delta Y_{\text{err}}| \quad (22)$$

$$J_c = \sum_{i=1}^n |a_y| \quad (23)$$

$$J_e = \sum_{i=1}^n |jerk_y| + \sum_{i=1}^n |a_y| \quad (24)$$

where J_t is the tracking performance indicator, ΔY_{err} is the lateral tracking error, a_y and $jerk_y$ are lateral acceleration and jerk, respectively. The smaller the indicators in the formula, the better is the tracking performance, comfort and economy. The index parameters are listed in table 9. The corresponding results are shown in figure 18.

In figure 18(a), the lateral tracking error based on the AHP-MPC is reduced by 35.28%, 9.62% and 29.32%, respectively, compared to the three comparison methods. This is because the AHP-MPC optimizes the HC based on optimization method under different vehicle speeds and curvatures, therefore, the EV can always obtain the HC with minimum tracking error. In figure 18(b), the comfort of the AHP-MPC-based method is increased by 19.53%, 5.49% and 15.16%, respectively, compared to the other three methods. In figure 18(c), the economy of the AHP-MPC method is improved by 21.84%, 5.42% and 12.55%, respectively, compared to the other three methods. The three algorithms compared do not accurately fit the relationship between HC, curvature and vehicle speed, and the optimization methods adopted also have certain limitations. Therefore, when the tracking error occurs, the controller cannot adjust in time, increasing the lateral acceleration, which reduces the comfort and economy of the EV.

7. Summary

In this study, to realize safe LC and obstacle avoidance of autonomous vehicles, an area discrimination method for safe LC based on the CSVTH was proposed considering the driving of the SVs. To improve

the path tracking performance, a WAPSO algorithm was proposed to optimize the HC of the MPC controller under different vehicle velocities and road curvatures, and the calculation methods of vehicle velocity, curvature and the optimal HC were further obtained. The results showed that the AHP-MPC could significantly increase the accuracy and computational efficiency of path tracking.

In future research, the following aspects should be considered. First, the sampling time of the controller can be optimized according to the vehicle speeds and curvatures to further improve the computational efficiency of the controller. Second, optimal control of the vehicle longitudinal speed is important for accurate path tracking. Third, future work will extend the approach to urban driving and multi-agent coordination under communication delays.

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Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

CRedit authorship contribution statement

Jun Chen: Conceptualization, Software, Writing—original draft. Fazhan Tao: Validation, and editing. Zhumu Fu: Methodology. Nan Wang: Writing, review.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

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